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1. Introduction

This study attempts to account for the restrictions on the movement of/from subjects in Chomsky's (2024) MC framework, which proposes a new viewpoint for movement by replacing successive-cyclic movement with an interpretive operation, Access. We discuss the long-standing asymmetries between subjects and objects concerning movement in this radically changed framework. We propose a syntactic operation, Mark, as a phase-level operation, demonstrating that the restriction of movement of subjects is derived from Mark and the concept of theta-domains. The remainder of this paper is organized as follows. Section 2 introduces the MC framework by Chomsky (2024). We focus on the problem of assigning theta-roles in interpretive systems. Section 3 proposes that a phase-level operation Mark designates a syntactic object to be accessed by the interpretive systems. Sections 4 and 5 discuss the *that*-trace effect and extraction from subjects within the Mark and Access framework. Section 6 presents the concluding remarks.

2. MC framework

Chomsky (2024) proposes a framework in which the theta/non-theta segregation is crucial. In this framework, Merge is defined in terms of theta-roles.

- (1) a. $\{_{XP} X WP\} \{_{YP} Y ZP\}$
b. $\{\{_{XP} X WP\}, \{_{YP} Y ZP\}_{\theta}\}$
- (2) a. $\{_{UP} U \{\dots\{_{YP} Y, ZP\}_{\theta}\}\}$
b. $\{_{YP} \{_{UP} U \{\dots\{_{YP} Y, ZP\}_{\theta}\}\}\}$

(1) shows an application of External Merge (EM), where YP receives a theta-role. Thus, the output of Merge is theta-related.¹ (2) is an instance of internal Merge (IM), which applies to syntactic objects in the theta-structure. In this case, the input of Merge is theta-related. Once IM-ed, the higher inscription of YP in (2b) is no longer relevant to theta-structures (Duality of Semantics). Thus, it cannot be a participant in further applications of Merge. Therefore, multiple applications of movement are not allowed, and there is no successive-cyclic movement. The effects of long-distance movement are derived from Access by a phase head.

- (3) a. which picture of each other did Mary think the men hope Bill likes ~~which pictures~~
~~of each other~~ (Chomsky (2024: 31))
b. $W_1 [_{PH1} C INFL [Mary thought [_{PH2} C INFL [the men [W_2 hope Bill [_{PH3} W_3 like W_4]]]]]]]$ (*ibid.*)
c. scope: PH_1

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¹ This argument must take the theta-structure to be an extended concept more than traditionally assumed. Functional heads are also introduced by EM, but C or T is not a head creating the traditional theta-structure.

externalization: PH₁
 anaphor: PH₂
 theta-role: base-generated position (PH₃)

In (3b), the positions of *W* show where the *wh* phrase can be interpreted in the interpretive systems. In the derivation, the *wh* operator moves from *W*₄ to *W*₃. Since it is dissociated from the theta-structure, it can no longer move. However, as *W*₃ is a phase edge, the phase heads can access the position to provide the interpretations, as shown in (3c). In the CI system, the theta-role is assigned in PH₃, and the phase head of PH₁ accesses *W*₃ to determine the scope position as the matrix clause. Additionally, the *wh* operator is externalized at the sentence-initial position in the SM system by the access by *W*₁. In contrast, the anaphoric interpretation between *the men* and *each other* in the *wh* operator requires the *wh* operator to be interpreted in PH₂. Chomsky assumes that not every position, but a phase edge, can be accessed. Therefore, syntactic objects should be located at a phase edge to receive a long-distance interpretation.

Kitahara (2024) refines Chomsky's framework. In his framework, Access does not require the movement to a phase edge; rather, movement out of the base-generated position is sufficient for Access. A difference is found in subject *wh* interrogatives.

- (4) a. Who does it seem ate the sandwich?
 b. C/Q does it seem [*who*₁ [INFL [*who*₂ [_{v*P} [ate the sandwich]]]]]]
 (Chomsky (2024: 34))

*Who*₂ in (4b) is the base-generated position, from where it moves to *who*₁ by normal subject raising. It cannot move further. Although neither *who*₁ nor *who*₂ is in a phase edge, Chomsky assumes that C accesses *who*₂. Since it lacks externalization features, its copy, *who*₁, is externalized at C/Q. This analysis faces two potential problems. The first one is that it allows (indirect) access to a non-phase-edge position, which weakens the original assumption that phase edges are the positions for Access. The second problem is pointed out by Kitahara (2024), which concerns the following example.

- (5) a. (guess) who you think was arrested
 b. [arrested *who*₁]
 c. [was [arrested *who*₁]]
 d. [Infl [was [arrested *who*₁]]]
 e. [*who*₂ [Infl [was [arrested *who*₁]]]]
 f. [C [*who*₂ [Infl [was [arrested *who*₁]]]]]
 (Kitahara (2024: 206))

In the MC framework, *who* stays at the position of *who*₂, the spec of TP. However, it is not a phase edge. Moreover, *who*₁ is also located in a non-phase-edge position, and therefore, the matrix phase head cannot access *who*₁ either.

In Kitahara's framework, *who*₂ can be accessed because it undergoes movement and exits from the theta-domain.

Leaving the details aside, both frameworks exploit the segregation between theta and non-theta positions. However, Chomsky (2024: 27) notes that using the notion of theta-roles in syntax produces a look-ahead problem, and it remains unclear how theta-positions are technically defined. In the next section, we discuss the role of syntax in theta-assignment.

3. Proposal

In this study, we define a theta-assignment mechanism using labeling. Since syntactic structures are interpreted according to labels (Chomsky 2015), we assume that syntax provides a domain for theta-assignment as follows.

- (6) The domain labeled v^* is the theta-domain.

Syntactic objects in the theta-domain are assigned theta-roles in the interpretive systems.² To interpret the items in the syntactic structures, we propose (2) and (3).

- (7) Mark: A local phase head marks visible syntactic objects in the theta-domain.³
(8) A marked element is accessed by a phase head in the interpretative systems.

(7) applies in syntax and specifies the syntactic object to be interpreted in interpretive systems. (8) corresponds to Chomsky's (2024) Access and provides the interpretation at the interpretive systems.

Unlike Chomsky's and Kitahara's frameworks, we propose that movement to a phase-edge/clausal domain is unnecessary. For Access, syntactic objects must be i. visible and ii. located in a theta-domain at the phase level. The derivation proceeds as follows.

- (9) a. Inside syntax: i. Merge and ii. Copy Formation
b. Phase-level operations: i. labeling, ii. determination of the theta-domains, and iii. Mark
c. In the interpretive systems: Access by phase heads

(9aii) follows Chomsky's (2024) assumption that copy relations are assigned between inscriptions with the c-command relation in the same phase. In phase-level operations, the theta-domains are determined as the domain labeled v^* , and a phase head marks the syntactic objects inside it. Finally, all the phase heads can access the marked syntactic objects for interpretation at the interpretive systems. Next, we propose that Mark must respect the following condition.

- (10) The label C closes the accessible domain for Mark.

(10) induces locality effects. The intuition lies in the division of labor in the phasal interpretations; the v^*P phase forms the theta-domain, and the CP phase expresses the discourse domain. These two phases complete a clause. Assuming that they constitute a minimal unit of human language, we propose that a syntactic object must be marked in this restricted domain. In effect, (10) can capture the main effect of the phase impenetrability condition—syntactic operations are restricted within the CP.

Although v^* can mark a syntactic object as a phase head, it does not have the same effect as (10). Thus, there is asymmetry between v^* and C, reflecting the fact that CP is required for a convergent clause, and the CP structure is a basic unit of human language.

Before presenting the analysis, two notes are in order. First, (10) is based on labeling, a phase-level operation in syntax. This implies that the accessible domain for Mark is determined dynamically during the course of derivation. Second, once marked by a local phase head, the element can be

² For the space limitation, we leave the exact mechanism of assigning theta-roles for future research. It depends on lexical properties how many theta-roles a verb assigns or what role is assigned to the highest argument.

³ We assume either phase head (v^* or C) can mark syntactic objects inside the theta-domain.

accessed in the interpretive systems by any phase head. Mark chooses the element to be assigned a discourse effect. The Markovian property is not problematic. It precludes the evaluation of the derivation based on the kind of operation applied to a syntactic object, but Mark is more like labeling or Agree, and assigns some properties to a syntactic object.

Comparing the proposed Mark and Access framework with Chomsky's and Kitahara's frameworks, all three systems require a mechanism that designates a syntactic object to be assigned interpretations for long-distance movement. The first one employs Mark, whereas the latter two utilize movement and mapping rules. The next section considers how our framework accounts for the movement phenomena.

4. Analysis

4.1. *That*-Trace Effect

Chomsky (2015) explains *that*-trace effect in the labeling theory. However, the analysis crucially depends on successive cyclic movement, which is abandoned in the MC framework. This study aims to explain this phenomenon in terms of (10). (12) shows the CP phase of (11).

- (11) Who do you think **(that)* loves Mary?
 (12) [C [who [INFL [v^* ~~who~~ [v^* loves Mary]]]]]]

Because of the subject movement, the underlined structure is labeled v^* and becomes a theta-domain. However, the inscription of *who* inside the theta-domain cannot be marked because it is invisible due to the lower copy status. The higher copy is visible, but not in the theta-domain. Thus, neither inscription can be marked. Subsequently, the topmost label of (12) becomes C, and Mark cannot examine this structure. This is why the subject in (11) cannot be interpreted (externalized) in the sentence-initial position. Next, (13) shows a part of the structure of (11) without *that*.

- (13) [think [who [INFL [v^* ~~who~~ [v^* loves Mary]]]]]]

We assume that without *that*, the embedded structure is IP (Bošković 1996). Thus, (10) does not apply here, and the accessible domain for Mark is extended.⁴ At a matrix phase, the structure in (13) is labeled v^* and constitutes the theta-domain. Thus, the higher inscription of *who* becomes visible for Mark in the matrix phase.

Note that Chomsky's and Kitahara's framework cannot explain the difference between (12) and (13). In Chomsky's analysis, C can access the base-generated position of the subject *wh* operator in (12), and indirect access enables the subject in the spec of TP to be externalized at a higher position. Kitahara's framework also allows access to the moved subject in the spec of TP in (12).

We have demonstrated that the proposed framework can accommodate the examples of the *that*-t effect. A further advantage of this analysis is that it can be extended to the general COMP-*t* effect.

- (14) Who would you prefer (**for*) *t* to meet Sue at the station? (Pesetsky (2017: 1000))

Although it remains unclear how the clause *for NP to...* is analysed in Chomsky's (2015) framework, (14) poses a potential problem for Chomsky (2015), as the label $\langle \phi, \phi \rangle$ seems irrelevant to the *for*-t effect. In our analysis, assuming that *for* also closes the domain for Mark as a phase head, (14) fails to mark the visible subject.

⁴ The same effect can be produced by C deletion (Chomsky 2015).

4.2. Extraction from Subjects

The analysis can naturally be extended to extraction from subjects. As is well-known, subjects form islands.

- (15) a. *Who did stories about terrify John? (Chomsky 1973: 249)
 b. Who did John hear stories about? (Chomsky 1973: 249)
- (16) [C [SUBJ [INFL [v* SUBJ [v* v* [OBJ [R, ~~OBJ~~]]]]]]]

(16) shows the CP structure of (15a, b). After movement to the spec of TP, the subject *wh* operator exits from the theta-domain. Thus, it cannot be accessed. Conversely, *v** merges at a position higher than that of the shifted object. Therefore, after movement, the higher inscription remains in the theta-domain and can be marked by a phase head.

Other languages show that extraction from in-situ subjects is possible—the restriction does not concern the notion of the subject, but extraction from the spec of TP is banned.

- (17) a. ??Dono hon-o Mary-ga [John-ga ~~dono hon-o~~] katta koto]-ga
 which book-ACC M.-NOM J.-NOM which book-ACC bought fact-NOM
 mondai-da-to omotteiru-no?
 problem-COP-C think-Q
 ‘Lit. Which book is it that Mary thinks that the fact that John bought it is a problem?’
 (Japanese; Lasnik and Saito 1992: 124, cf. Ross 1967: 263)
- b. ¿De qué equipo dices que han bailado [dos participantes ~~de qué equipo~~]?
 of what team say that have danced two participants of what team
 ‘Which team do you say that two members of have danced?’
 (Spanish; Gallego and Uriagereka 2007: 57)

Lasnik and Saito (1992) observe that (17a) is not worse than the extraction from objects, concluding that there is no subject island in Japanese. As Japanese allows in-situ subjects (Kuroda 1988, Nishioka 2019), the subject in (17a) can be in the base position. Additionally, nominals with case markers become invisible in labeling (Saito 2016, Hayashi-Katsu 2025). Thus, in the structure {SUBJ, *v**P}, *v** can take the label, and the in-situ subject can be located in the theta-domain. Therefore, Japanese allows extraction from subjects because they can stay in the base-generated positions. In the Spanish example (17b), V precedes the bracketed in-situ subject. The order is derived by V-to-INFL raising, by which the domain labeled *v** is extended to the INFL level. Hence, the in-situ subject in (17b) is also located in the theta-domain, accessible to Mark.

5. Complicated Issues

5.1. Subject *Wh* Interrogatives

In this framework, subject *wh* clauses are analyzed differently from other *wh* interrogatives.

- (18) a. Who loves Mary?
 b. [who [INFL [~~who~~ [v* v* loves Mary]]]]

According to the vacuous movement hypothesis (Chomsky 1986, 2015), the *wh* subject only moves to the spec of TP, not to the spec of CP. However, the position is not in the theta-domain. Thus, neither inscription is a candidate for Mark, as shown in the example of the *that*-trace effect. Nevertheless, the sentence is of course grammatical.

I argue that the matrix subject *wh* interrogative clause does not require Access or Mark. Access is required to capture the long-distance movement in the MC framework, in which the phase heads designate where a *wh* operator takes its scope and where it is externalized. In the case of the matrix *wh* subject interrogatives, INFL has the [Q] feature and the [phi] features by virtue of feature inheritance (Chomsky 2015). Therefore, the label <Q, Q> provided by INFL can designate the position for interpretation of the subject *wh* operator (Hayashi 2022).

5.2. Passives

Chomsky (2008) observes that the subjects of passives behave like objects regarding extraction.

- (19) a. Of which car did [they find the (driver, picture)]?
 b. *Of which car did [the (driver, picture) cause a scandal]?
 c. Of which car was [the (driver, picture) awarded a prize]?

(Chomsky 2008: 147)

(19a, b) show extraction from an object and a subject, respectively. (19c) represents a well-formed extraction from the subject of passives. According to Epstein, Kitahara and Seely (2016) and Hayashi (2020), the construction involves the R-*v** amalgam derived by external pair-Merge. In (20), NP represents the subject of passives.

- (20) [C [NP [INFL [_{R-v*} R-*v**, NP]]]]

Although NP moves to the spec of TP, and the lower inscription becomes invisible after the movement, we assume that R-*v** acts like a phase head (akin to weak phase in Chomsky 2008) and can mark the subject at the R-*v**P level. In the CP phase, the subject moves to the spec of TP, but the lower inscription has already been marked. Therefore, phase heads can access (the structure inside) NP in the interpretive systems.

5.3. On the Copular Constructions

In addition to passives, Chomsky (2008) observes that extraction from a subject is also possible if the subject undergoes the successive cyclic A movement.

- (21) Of which car did [the (driver, picture) likely [~~the (driver, picture) of which car~~ to [the (driver, picture) of which car cause a scandal]]]? (Chomsky 2008: 153)

(21) faces two problems: i. as the verb is transitive and the subject is in the spec of *v**P, the solution presented in the passive case does not work; and ii. successive-cyclic A movement is not assumed by Chomsky (2021, 2024). We assume that the raising predicate (*likely* in (21)) acts like a (weak) phase head and marks the subject. Chomsky (2008) assumes an intermediate copy, to which he attributes the grammaticality, but we argue with Chomsky (2021, 2024) that the subject directly moves from the base position to the matrix spec of TP. Well-formedness results from the (weak) phase status of the raising predicate.

Without such a predicate, the subject of normal copular clauses does not allow extraction.

- (22) *the man who [pictures of ~~who~~] are on the table. (Chomsky 1986: 31)

The ill-formedness is due to the lack of a predicate that serves as a weak head and can mark the subject.

Next, let us consider the following contrast, which poses a potential problem for the analysis so far.

- (23) a. *Which candidate were [posters of ~~which candidate~~] all over town? (Merchant 2001: 187)
 b. Which candidate were there [posters of ~~which candidate~~] all over town? (Merchant 2001: 187)

Contrary to (22) and (23a), the logical subject of the there-construction in (23b) allows extraction without a predicate other than copular. Although the detailed explanation depends on the exact structure of the *there*-construction, where a labeling problem occurs between the logical subject and a locative expression, one possible solution for this XP-YP problem is to assume a functional head above the subject, which is a candidate for a head that marks the subject.

5.4. Multiple *Wh* Interrogatives

We have discussed that the *that*-t effect and the subject island effect result from the failure of Mark. This section considers how our analysis accommodates other island effects.

Hayashi (2025) argues that some constraints on movement do not restrict movement/access itself, but they are actually constraints for reconstruction. Thus, the complex NP island and the superiority effect ban overt movement in (24), but the covert counterparts are well-formed, as shown in (25).

- (24) a. *What did Ken hear [the rumor that Mary bought ~~what~~]
 b. *What did who buy?
 (25) a. Who is reading a book that criticizes who? (Watanabe (2001: 207))
 b. Who remembers where we bought what? (Watanabe (2001: 207))

Regarding this point, English behaves as an in-situ language like Japanese.

- (26) Ken-wa [Mary-ga nani-o kat-ta uwasa]-o kii-ta no?
 K.-TOP M.-NOM what-ACC buy-PST rumor-ACC hear-PST Q
 ‘What did Ken hear [the rumor that Mary bought]?’ (Hayashi (2025: 15))

Although they are externalized in the base-generated position, we assume that in-situ *wh* operators in English should also be marked in syntax. Because the object *wh* operators in (25) are located in theta-domains, they can be marked. The scopes are determined by Access, but the SM system does not require them to be externalized at the scope positions in multiple *wh* interrogatives. Since there is no reconstruction, they evade constraints inducing island effects. Next, consider the following example.

- (27) *We’re trying to find out which man said that which woman was in love with him. (Kayne (1983: 234))

(27) shows that an in-situ *wh* operator induces the *that*-t effect. This behavior is not expected in Chomsky’s (2015) framework since *which woman* in (27) can provide the label $\langle \phi, \phi \rangle$. In our analysis, the constraint on the *that*-t effect concerns the syntactic operation Mark, unlike island constraints observed in (25). Thus, whether or not the *wh* operator is externalized at another position, (27) becomes ill-formed due to the failure of Mark for *which woman*.⁵

⁵ Kayne observes that additional *wh* operator improve (27). We leave this issue open for future

6. Concluding Remarks

This paper has started with the discussion of the role of syntax in assigning theta-roles. In the proposed framework, Chomsky's (2024) Access is decomposed into a syntactic part and a part in the interpretive systems. For the former property, we propose Mark as a phase-level operation. Mark is possible in a domain with the label v^* , which is responsible for the theta-assignment. The CP phase closes the Mark domain and completes a clause. Thus, this framework further pursues Chomsky's (2024) intuition that the bifurcation between the predicate/discourse structures forms fundamental properties in human language. Although the proposed analysis is based on Chomsky's (2024) and Kitahara's (2024) frameworks, all three systems are different in which property of human language is used for Access, which is summarized in (28).

- (28) The crucial distinction for Access
- a. The proposed analysis: theta-domain or not
 - b. Chomsky (2024): phase edge or not
 - c. Kitahara (2024): base-generated position or not

In the proposed framework, we have analyzed the *that*-trace effect and the subject island condition using Mark. We have also considered three cases showing interesting behaviors: the subjects of passives, raising predicates, and copulars. Finally, we have discussed the derivation of multiple *wh* operators. If our analysis is on the right track, constraints on movement are divided into two types: the first type is actually a constraint on Mark in syntax, and the second type precludes reconstruction across islands.

References

- Bošković, Željko (1996) "Selection of Infinitival Complements," *Natural Language and Linguistic Theory* 14, 269-304.
- Chomsky, Noam (1973) "Conditions on Transformations," *A Festschrift for Morris Halle*, ed. by Stephen R. Anderson and Paul Kiparsky, 232-286, Holt, Rinehart and Winston, New York.
- Chomsky, Noam (1986) *Barriers*, MIT Press, Cambridge, MA.
- Chomsky, Noam (2008) "On Phases," *Foundational Issues in Linguistic Theory: Essays in Honor of Jean-Roger Vergnaud*, ed. by Robert Freidin, Carlos Otero, and Maria-Luisa Zubizarreta, 133-166, MIT Press, Cambridge, Massachusetts.
- Chomsky, Noam (2015) "Problems of Projection: Extensions," *Structures, Strategies and Beyond: Studies in Honour of Adriana Belletti*, ed. by di Domenico, Elisa, Cornelia Hamann and Simon Matteini, 3-16, John Benjamins, Amsterdam/Philadelphia.
- Chomsky, Noam (2021) "Minimalism: Where Are We Now, and Where Can We Hope to Go," *Gengo Kenkyu* 160, 1-41.
- Chomsky, Noam (2024) "The Miracle Creed and SMT," *A Cartesian Dream: A Geometrical Account of Syntax. In Honor of Andrea Moro*, ed. by Matteo Greco and Davide Mocci, 17-39, Lingbuzz Press.
- Epstein, Samuel David, Hisatsugu Kitahara and T. Daniel Seely (2016) "Phase Cancellation by

research.

- (i) *We're trying to find out which man said that which woman was in love with which boy.
(Kayne (1983: 234))

- External Pair-Merge of Heads,” *Linguistic Review* 33, 87–102.
- Gallego, Ángel J. and Juan Uriagereka (2007) “Condition on Sub-Extraction,” *Coreference, Modality, and Focus*, ed. by Luis Eguren and Olga Fernández Soriano, 45-70, John Benjamins, Amsterdam/Philadelphia.
- Hayashi, Norimasa (2020) “Labeling without Weak Heads,” *Syntax* 23, 275-294.
- Hayashi, Norimasa (2022) *Labels at the Interfaces: On the Notions and the Consequences of Merge and Contain*, Kyushu University Press, Fukuoka.
- Hayashi, Norimasa (2025) “Notes on Variations of Wh Constructions in the MC Framework and the Principle T Feeding Reconstruction,” *Nanzan Linguistics* 19, 9-20.
- Hayashi-Katsu, Norimasa (2025) *Case as a Lower Copy: Scrambling in Japanese Based on a Form Copy and Labeling Framework*, ms. Nanzan University.
- Lasnik, Howard and Mamoru Saito (1992) *Move α* , MIT Press, Cambridge, MA.
- Merchant, Jason (2001) *The Syntax of Silence: Sluicing, Islands, and the Theory of Ellipsis*, Oxford University Press, Oxford.
- Nishioka, Nobuaki. (2019) “Discourse-Configurationality and the Scope of Negation,” *Nanzan Linguistics* 14, 25-55.
- Kayne, Richard (1983) “Connectedness,” *Linguistic Inquiry* 14, 223-249.
- Kitahara, Hisatsugu (2024) “A Review and Reanalysis of Some Recent Developments from GK to MC, and beyond,” *Reports of the Keio Institute of Cultural and Linguistic Studies* 55, 199-214.
- Kuroda, S.-Y. (1988) “Whether Agree or Not,” *Papers from the Second International Workshop on Japanese Syntax*, ed. by William J. Poser, 1-47, CSLI, Stanford.
- Pesetsky, David Michael (2017) “Complementizer-Trace Effects,” *The Blackwell Companion to Syntax: Second Edition*, ed. by Martin Everaert and Henk C. van Riemsdijk, 993-1026, Hoboken, John Wiley & Sons, Inc.
- Ross, John Robert (1967) *Constraints on Variables in Syntax*, Doctoral dissertation, MIT.
- Saito, Mamoru (2016) “(A) Case for Labeling: Labeling in Languages without ϕ -Feature Agreement,” *The Linguistic Review* 33, 129–175.
- Watanabe, Akira (2001) “Wh-in-situ Languages,” *The Handbook of Contemporary Syntactic Theory*, ed. by Mark Baltin and Chris Collins, 203-225, Malden, MA and Oxford, Blackwell Publishers.